Uncertainty-Aware Open-Vocabulary Semantic Gaussian Splatting for Robust Mapping in Dynamic Scenes

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Abstract—Semantic 3D Gaussian Splatting (3D-GS) enables open-vocabulary scene understanding by embedding language features into a photorealistic map, but existing representations report semantics without a calibrated notion of trust. This is unsafe for robotics: a region that has been observed only briefly is treated identically to one whose semantics are genuinely ambiguous or changing. We present a probabilistic open-vocabulary semantic field for 3D-GS that attaches a per-Gaussian directional posterior to the map and, from the same sufficient statistics, exposes a disentangled epistemic and aleatoric uncertainty. Because language embeddings are unit-normalized, we model each Gaussian's semantics with a von Mises–Fisher (vMF) distribution and derive a closed-form Bayesian Kernel Inference (BKI) update that is the continuous generalization of categorical semantic mapping. We further show that the required kernel-weighted sufficient statistics are recoverable as the gradient of a feature render, so the update plugs into any differentiable feature rasterizer without modifying its kernels. The disentangled uncertainty drives an uncertainty-driven consolidation rule that identifies stable structure and forgets transient content, yielding robustness to scene change without hand-tuned dynamic-object heuristics or a language model in the loop. We evaluate calibration, open-vocabulary segmentation, and localization under scene change on indoor benchmarks. [TODO: One-sentence headline result once measured.]

Index Terms—Mapping, semantic scene understanding, localization, RGB-D perception, uncertainty estimation.

I. Introduction

RELIABLE spatial perception underpins autonomous operation in complex indoor environments, where robots must localize and act despite occlusions, repetitive structure, diverse object categories, and dynamic rearrangement of the scene [22]. Recent work couples photorealistic 3D Gaussian Splatting (3D-GS) [1] with language features [3] to build open-vocabulary semantic maps that support querying arbitrary concepts at test time [17]–[19]. Such maps are attractive for localization and navigation because they combine appearance, geometry, and semantics in a single differentiable representation.

A critical property is missing, however. Existing semantic 3D-GS representations report a most-likely semantic label or feature at every point but provide no calibrated measure of how much that report should be trusted. For a safety-critical robot this conflation is dangerous: a surface seen from only one fleeting viewpoint and a surface whose semantics are genuinely ambiguous or actively changing are reported with identical confidence, even though the correct response to each is opposite—gather more observations in the first case, distrust the region in the second. Continuous Semantic Splatting (CSS) [4] took an important step by porting Bayesian Kernel Inference (BKI) [5], [6] from voxel maps to 3D-GS, obtaining a Dirichlet posterior over a closed label set. Two gaps remain. First, open-vocabulary semantics are continuous CLIP directions, not categories, so a Dirichlet treatment does not apply; CSS itself identifies an open-dictionary continuous conjugate prior as future work. Second, a single posterior variance does not separate uncertainty that more sensing can reduce (epistemic) from uncertainty that it cannot (aleatoric), yet that distinction is exactly what a robot needs to decide whether to look again or to forget.

We address both gaps with a probabilistic open-vocabulary semantic field for 3D-GS. Our representation is summarized in Fig. 1. The contributions are:

- A directional semantic posterior for 3D-GS that models each Gaussian's open-vocabulary semantics with a von Mises–Fisher distribution on the unit hypersphere and admits a closed-form BKI update—the continuous generalization of categorical semantic mapping, using the splat weight as the inference kernel.
- A disentangled epistemic/aleatoric read-out obtained from the same sufficient statistics at no extra cost, with an explicit small-sample correction that prevents the false confidence that arises from few observations.
- A gradient-based accumulation showing that the kernel-weighted sufficient statistics equal the gradient of a feature render, so the update integrates with any differentiable feature rasterizer without custom kernels.
- An uncertainty-driven consolidation mechanism that uses the aleatoric signal to identify stable structure and forget transient content, providing robustness to scene change without dynamic-object thresholds or a language model in the loop.

We evaluate calibration, open-vocabulary segmentation, and localization under scene change on standard indoor datasets, and ablate each component.

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II. Related Work

Visual localization and rendering-based refinement. Absolute pose regression [20] and scene coordinate regression [21] learn a mapping from images to pose but generalize poorly across scenes and degrade in complex layouts. Rendering-based methods refine a coarse pose against a learned scene representation; replacing NeRF [2] with 3D-GS [1] improves render quality and efficiency for relocalization. [22] augment a Gaussian map with semantics and a language model to select stable objects for robust indoor localization. We share the goal of robustness to scene change but replace heuristic stable-object selection with a calibrated uncertainty derived directly from the map.

Semantic and language-embedded 3D-GS. A growing body of work distills 2D features into a 3D-GS field for open-vocabulary understanding [17]–[19]. These methods optimize or rasterize per-Gaussian features but treat the result as deterministic, providing no uncertainty for downstream decision making. Bayesian Fields [9] fuses semantics across views with a recursive Bayesian update, but maintains only a scalar task-relevance probability per Gaussian for a query fixed at mapping time, and assumes a static scene. Our field instead maintains a posterior over the embedding direction itself, so arbitrary queries can be answered post hoc with calibrated uncertainty.

Uncertainty in mapping. BKI [5] provides closed-form continuous mapping with quantifiable uncertainty and underlies several semantic mapping systems [6]. Evidential extensions move beyond one-hot labels [7]. CSS [4] brings BKI to 3D-GS with a Dirichlet posterior over a closed label set, and concurrent work [8] infers physical properties with a Bayesian 3D-GS treatment that separates aleatoric and epistemic terms. We extend the BKI line to open-vocabulary semantics via a directional (vMF) conjugate model and provide the epistemic/aleatoric split for continuous embeddings.

Directional statistics for embeddings. Normalized embeddings lie on a hypersphere, where the vMF distribution is the natural model and its concentration measures confidence [12], [13]. Concurrent work uses a vMF model of language-embedding uncertainty in an implicit field for navigation [10]; we instead attach the directional posterior to an explicit 3D-GS map, derive its closed-form BKI update, and disentangle epistemic from aleatoric uncertainty.

Active perception. Active inference casts perception and action as minimizing expected free energy, which decomposes into information-seeking (epistemic) and goal-directed (pragmatic) value [14], [15]. Recent semantic 3D-GS exploration combines geometric and semantic information gain with hand-set weights [11]. Our disentangled uncertainty provides the epistemic term in a principled form and, by gating out aleatoric regions, avoids expending observations where more sensing cannot help (Sec. III-E).

III. Method

A. Directional Semantic Field

We build on an open-vocabulary 3D-GS map of N Gaussians $\{g_n\}_{n=1}^{N}$, each with mean, covariance, opacity and color as usual [1]. We attach to each Gaussian a semantic state living on the unit hypersphere $\mathbb{S}^{d-1}$, matching the geometry of ℓ_2-normalized language embeddings [3].

Let a training/keyframe provide a dense per-pixel CLIP feature map; pixel p contributes a unit observation $\mathbf{z}_p \in \mathbb{S}^{d-1}$. We model the observations of Gaussian n with a von Mises–Fisher (vMF) likelihood with fixed concentration κ,

$$p(\mathbf{z}_p \mid \boldsymbol{\mu}_n) = C_d(\kappa)\, \exp\big(\kappa\, \boldsymbol{\mu}_n^\top \mathbf{z}_p\big), \qquad \boldsymbol{\mu}_n \in \mathbb{S}^{d-1}, \tag{1}$$

where $\boldsymbol{\mu}_n$ is the (unknown) mean semantic direction and $C_d(\kappa)$ the normalizer. The conjugate prior for the mean direction of a vMF likelihood with fixed κ is again a vMF, $\boldsymbol{\mu}_n \sim \mathrm{vMF}(\boldsymbol{\mu}_0, \kappa_0)$ [12]; setting $\kappa_0 = 0$ recovers a uniform prior, while a nonzero κ_0 lets a CLIP text embedding act as a semantic prior.

Following BKI [5], we relate observations across space by tempering each likelihood with a kernel $k_{p,n} \in [0, 1]$ that measures the influence of pixel p on Gaussian n. As in CSS [4], the 3D-GS blending weight is a valid BKI kernel: $k_{p,n} = T_{p,n}\, \alpha_{p,n}$, the transmittance-weighted opacity with which g_n contributes to p. The tempered posterior is

$$p(\boldsymbol{\mu}_n \mid \mathcal{D}) \propto \exp(\kappa_0 \boldsymbol{\mu}_0^\top \boldsymbol{\mu}_n) \prod_p \left[C_d(\kappa) \exp(\kappa\, \mathbf{z}_p^\top \boldsymbol{\mu}_n)\right]^{k_{p,n}}. \tag{2}$$

Crucially $C_d(\kappa)^{k_{p,n}}$ is independent of $\boldsymbol{\mu}_n$ and drops out, leaving a vMF posterior in closed form:

$$p(\boldsymbol{\mu}_n \mid \mathcal{D}) \propto \exp\big(\boldsymbol{\eta}_n^\top \boldsymbol{\mu}_n\big) = \mathrm{vMF}\Big(\boldsymbol{\mu}_n \,\Big|\, \tfrac{\boldsymbol{\eta}_n}{\|\boldsymbol{\eta}_n\|}, \|\boldsymbol{\eta}_n\|\Big), \tag{3}$$

$$\boldsymbol{\eta}_n = \kappa_0 \boldsymbol{\mu}_0 + \kappa\, \mathbf{R}_n, \qquad \mathbf{R}_n = \textstyle\sum_p k_{p,n}\, \mathbf{z}_p, \tag{4}$$

with effective observation mass $n_n^{\mathrm{eff}} = \sum_p k_{p,n}$. The posterior mean direction and concentration are $\boldsymbol{\mu}_n^\star = \boldsymbol{\eta}_n / \|\boldsymbol{\eta}_n\|$ and $\kappa_n^\star = \|\boldsymbol{\eta}_n\|$, and the posterior angular variance admits the closed form $\approx (d-1)/\kappa_n^\star$. Equation (4) is the exact analogue of the categorical update $\alpha_n \leftarrow \alpha_0 + \sum_p k_{p,n}\, \mathbf{y}_p$ of [4] with the one-hot label $\mathbf{y}_p$ replaced by the continuous direction $\mathbf{z}_p$; the resultant vector $\mathbf{R}_n$ is the sufficient statistic, and its length relative to the accumulated mass separates agreement from conflict (Fig. 2). Semantic queries reduce to cosine similarity between $\boldsymbol{\mu}_n^\star$ and a query text embedding, and a novel-view semantic render with per-pixel uncertainty follows the probabilistic rasterization of [4].

B. Gradient-Based Accumulation

Computing $\mathbf{R}_n$ and n_n^{eff} appears to require, for every Gaussian, the list of pixels it touches and their weights—information that standard rasterizers do not expose. We avoid any kernel modification by observing that feature compositing is linear in the per-Gaussian feature. Let $\mathbf{f}_n$ be a per-Gaussian feature buffer and $F_p = \sum_n k_{p,n} \mathbf{f}_n$ the composited feature at p. Then

$$\frac{\partial}{\partial \mathbf{f}_n}\Big[\textstyle\sum_p \mathbf{z}_p^\top F_p\Big] = \textstyle\sum_p k_{p,n}\, \mathbf{z}_p = \mathbf{R}_n, \tag{5}$$

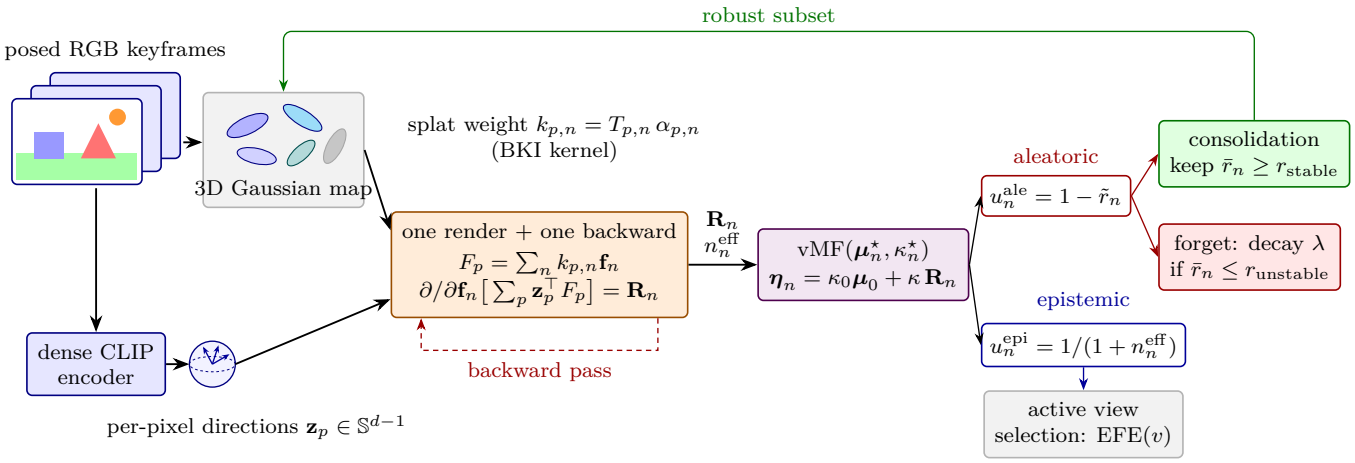


Fig. 1. Overview. Per-pixel CLIP directions $\mathbf{z}_p$ are accumulated into each Gaussian's von Mises–Fisher sufficient statistics $(\mathbf{R}_n, n_n^{\text{eff}})$ through the splat weight $k_{p,n} = T_{p,n}\alpha_{p,n}$ (the BKI kernel), recovered exactly by one feature render and one backward pass (Sec. III-B). The closed-form posterior yields a mean semantic direction and concentration, from which a disentangled epistemic/aleatoric uncertainty is read off (Sec. III-C) and used for consolidation (Sec. III-D) and, optionally, active viewpoint selection (Sec. III-E).

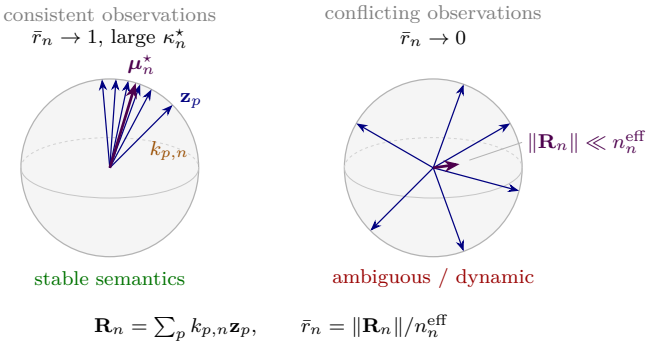


Fig. 2. Directional semantic state on $\mathbb{S}^{d-1}$. Kernel-weighted CLIP directions accumulate into the resultant $\mathbf{R}_n$, whose direction is the posterior semantics and whose length relative to the observation mass $(\bar{r}_n)$ separates consistent evidence (left) from conflicting evidence (right).

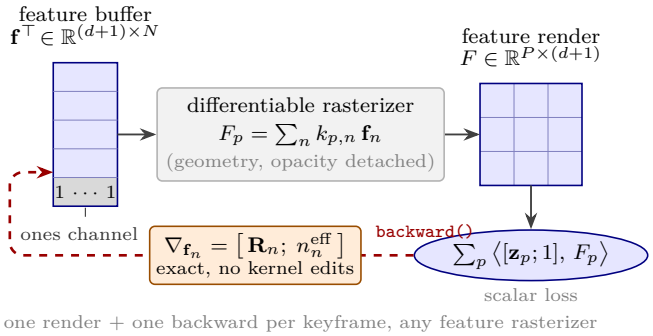


Fig. 3. Gradient-based accumulation. Because feature compositing is linear in the per-Gaussian buffer, the kernel-weighted sufficient statistics are the gradient of a single feature render; a constant ones channel recovers n_n^{eff} in the same backward pass, with no rasterizer modification.

i.e. the resultant is the gradient of a single feature render with respect to the feature buffer, with geometry and opacity detached so the weights are constant. Appending a constant ones-channel recovers n_n^{eff} in the same backward pass. Hence the update requires one render and one backward call through any differentiable feature rasterizer [16], [18]; Fig. 3 summarizes the construction. We verified this identity against an explicit weighted-sum reference and obtained agreement to numerical precision.

C. Disentangling Epistemic and Aleatoric Uncertainty

Define the mean resultant length $\bar{r}_n = \|\mathbf{R}_n\|/n_n^{\text{eff}} \in [0,1]$, the standard vMF concentration statistic [13]. With a weak prior the posterior concentration factorizes as

$$\kappa_n^\star \;\approx\; \kappa\, n_n^{\text{eff}}\, \bar{r}_n, \tag{6}$$

exposing two independent routes to a diffuse posterior. A small n_n^{eff} means few observations: the posterior is uncertain only for lack of data, and additional viewpoints reduce it. This is epistemic uncertainty, for which we use the monotone proxy

$$u_n^{\text{epi}} = \frac{1}{1+n_n^{\text{eff}}}. \tag{7}$$

A small $\bar{r}_n$ despite large n_n^{eff} means many observations that disagree—the language directions at that location point different ways—which more sensing under the same conditions cannot resolve. This is aleatoric uncertainty,

$$u_n^{\text{ale}} = 1 - \tilde{r}_n, \qquad \tilde{r}_n = \bar{r}_n - \frac{1-\bar{r}_n}{n_n^{\text{eff}}+1}, \tag{8}$$

where $\tilde{r}_n$ is a small-sample shrinkage of $\bar{r}_n$. The correction is essential: with a single observation $\bar{r}_n = 1$ trivially, which would falsely signal certainty; we therefore treat u_n^{ale} as valid only once $n_n^{\text{eff}} \geq \tau_{\min}$ and report it as undefined otherwise. Maintaining the pair $(n_n^{\text{eff}}, \bar{r}_n)$ rather than a single scalar is what makes the two regimes separable; Fig. 4 maps the regimes and the consolidation rule of Sec. III-D onto this plane. A data-driven per-Gaussian

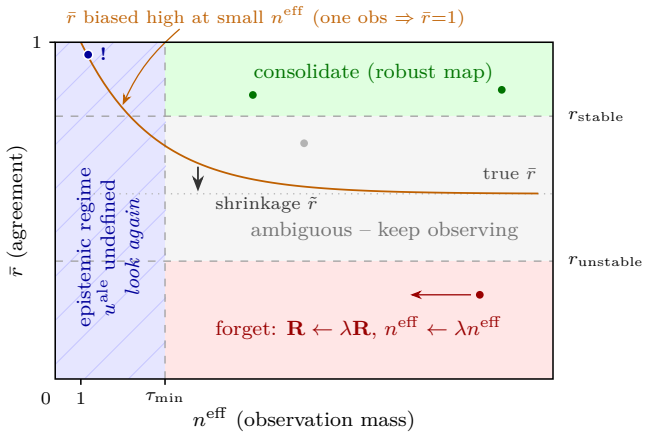


Fig. 4. Uncertainty regimes in the $(n^{\text{eff}}, \bar{r})$ plane. Low observation mass is the epistemic regime (more sensing helps; $\bar{r}$ is biased high and u^{ale} is withheld); given sufficient mass, high agreement consolidates into the robust map while persistent conflict is forgotten by evidence decay.

concentration estimate $\hat{\kappa}_n = \bar{r}_n(d - \bar{r}_n^2)/(1 - \bar{r}_n^2)$ [13] is available if a fixed κ is undesirable, at the cost of closed-form conjugacy.

D. Uncertainty-Driven Consolidation

The aleatoric signal identifies, without semantic thresholds or a language model, which structure is stable. Across keyframes, a Gaussian whose observations persistently disagree ($n_n^{\text{eff}} \geq \tau_{\text{min}}$, $\bar{r}_n \leq r_{\text{unstable}}$) corresponds to content that has moved or whose appearance is inconsistent; we decay its accumulated evidence, $\mathbf{R}_n \leftarrow \lambda\mathbf{R}_n$, $n_n^{\text{eff}} \leftarrow \lambda n_n^{\text{eff}}$ with $\lambda < 1$, an exponential forgetting of stale support. Conversely, Gaussians with $\bar{r}_n \geq r_{\text{stable}}$ are consolidated and define a robust subset used for localization. This replaces heuristic dynamic-object removal and stable-object selection with a single mechanism driven by the map's own uncertainty.

E. Active Viewpoint Selection (Optional)

The disentangled uncertainty also yields a principled next-best-view objective. Casting selection as minimizing expected free energy [14], a candidate view v with predicted per-Gaussian visibility $\nu_{v,n}$ is scored by an epistemic term that seeks reducible uncertainty and a pragmatic term that seeks task-relevant content:

$$\text{EFE}(v) = -\Big[\underbrace{\sum_n \nu_{v,n}\, u_n^{\text{epi}}(1 - u_n^{\text{ale}})}_{\text{epistemic value}} + \lambda_p \underbrace{\sum_n \nu_{v,n}\, s_n}_{\text{pragmatic value}}\Big], \tag{9}$$

where $s_n = \max(0, \boldsymbol{\mu}_n^{\star\top}\mathbf{t})$ is similarity to a query embedding $\mathbf{t}$. Gating the epistemic term by $(1 - u_n^{\text{ale}})$ prevents the agent from repeatedly visiting genuinely ambiguous or dynamic regions, a failure mode of objectives that do not separate the two uncertainties.

TABLE I
Open-vocabulary segmentation and calibration. OV: supports open-vocabulary queries. [TODO: replace placeholders; arrows indicate better direction.]

Method	OV	mIoU↑	Acc.↑	AUSE_{epi}↓	ECE↓
Deterministic feat. [18]	✓	XX.X	XX.X	–	–
CSS (Dirichlet) [4]	×	XX.X	XX.X	XX.X	XX.X
Ours (vMF)	✓	XX.X	XX.X	XX.X	XX.X

IV. Experiments

[TODO: All numeric cells below are placeholders (XX.X / –). Replace with your measured results before submission.]

A. Setup

Datasets. We evaluate on small-scale (7-Scenes [23]), large-scale (ScanNet [24]), and semi-dynamic indoor data; the dynamic split contains query images captured after object rearrangement [TODO: describe your custom dynamic dataset / acquisition].

Baselines. For open-vocabulary semantics with uncertainty we compare against the categorical Dirichlet treatment of CSS [4] and deterministic language-embedded 3D-GS [17]–[19]. For localization we compare against representative APR [20], SCR [21], and GS-based relocalization methods, including the semantic localizer of [22]. [TODO: finalize baseline list and citations.]

Metrics. Open-vocabulary segmentation accuracy / mIoU; calibration via the sparsification error (AUSE) for each uncertainty channel and expected calibration error (ECE); localization by median translation/rotation error (ATE/ARE) and success rate at 5 cm/5° and 2 cm/2°; and runtime. We report AUSE separately for the epistemic and aleatoric channels to test whether each predicts the failure mode it is intended to.

Implementation. [TODO: backbone for dense CLIP features (e.g. LSeg / MaskCLIP / SAM+CLIP), feature dimension d, κ, τ_{min}, r_{stable}, r_{unstable}, λ, hardware, timings.]

B. Open-Vocabulary Semantics and Calibration

Table I reports segmentation quality and calibration. Fig. 5 shows qualitative posterior semantics with epistemic and aleatoric maps. [TODO: 1–2 sentences interpreting the measured numbers: e.g. that the directional posterior matches/exceeds deterministic features on accuracy while providing calibrated uncertainty, and that the epistemic channel attains lower AUSE against reconstruction-improvable error while the aleatoric channel attains lower AUSE against dynamic/boundary error.]

C. Localization Under Scene Change

Table II reports localization accuracy. The consolidation mechanism (Sec. III-D) supplies a robust subset for matching under rearrangement. [TODO: interpret: e.g. that aleatoric-driven consolidation preserves accuracy as the scene changes, versus baselines that degrade.]

[TODO: qualitative figure: query RGB, posterior semantics, epistemic map, aleatoric map; show dynamic objects lighting up the aleatoric channel.]

Fig. 5. Posterior semantics with disentangled uncertainty. [TODO: caption.]

TABLE II
Localization on dynamic indoor data (median ATE/ARE and success rate). [TODO: replace placeholders.]

Method	ATE [cm] ↓	ARE [°] ↓	Success [%] ↑	
			5cm/5°	2cm/2°
APR [20]	XX.X	XX.X	XX.X	XX.X
SCR [21]	XX.X	XX.X	XX.X	XX.X
Semantic GS [22]	XX.X	XX.X	XX.X	XX.X
Ours	XX.X	XX.X	XX.X	XX.X

D. Ablations

Table III isolates each component. [TODO: interpret the four ablations: (i) directional vMF vs. categorical posterior on open-vocabulary queries; (ii) the 2-D $(n^{\text{eff}}, \bar{r})$ uncertainty with the small-sample correction vs. a single scalar; (iii) consolidation on/off under scene change; (iv) the EFE objective vs. a hand-weighted geometric+semantic score.]

TABLE III
Ablation study. [TODO: replace placeholders.]

Configuration	mIoU ↑	AUSE_{ale} ↓	ATE [cm] ↓
Categorical posterior	XX.X	XX.X	XX.X
Scalar uncertainty	XX.X	XX.X	XX.X
w/o consolidation	XX.X	XX.X	XX.X
Heuristic view score	XX.X	XX.X	XX.X
Full model	XX.X	XX.X	XX.X

V. Conclusion

We presented a probabilistic open-vocabulary semantic field for 3D Gaussian Splatting that attaches a von Mises–Fisher posterior to each Gaussian, derives a closed-form BKI update that generalizes categorical semantic mapping to continuous language embeddings, and exposes a disentangled epistemic/aleatoric uncertainty from the same sufficient statistics. The update integrates with any differentiable feature rasterizer through a gradient identity, and the uncertainty drives a consolidation rule that yields robustness to scene change without dynamic-object heuristics. [TODO: one sentence summarizing the strongest measured outcome.] Limitations include the fixed-concentration assumption and the dependence of the consolidation thresholds on a validation split; learning the concentration and an online open-set treatment are natural extensions.

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[TODO: Acknowledgment text.]

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